



University Institute of Engineering
Department of Computer Science & Engineering

Experiment-11

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1. Aim of the practical: To study and implement **Window Functions in PostgreSQL** to analyze employee salary history and identify salary patterns and top salary levels within departments.

2. Questions

(A) A company stores the monthly salary history of its employees in an EmployeeSalaryHistory table.

Write an SQL query to find the employees whose salary has **increased in every recorded month compared to the immediately previous month**.

For each salary record, compare the current month's salary with the employee's previous month's salary using a Window Function. Display only those employees for whom all later salary records are greater than their respective previous salary.

The output should display:

- Employee ID
- Employee Name

record_id	employee_id	employee_name	salary_month	salary
1	101	Alok	2026-01-01	50000
2	101	Alok	2026-02-01	52000
3	101	Alok	2026-03-01	55000
4	102	Priya	2026-01-01	60000
5	102	Priya	2026-02-01	58000
6	102	Priya	2026-03-01	62000
7	103	Raj	2026-01-01	45000
8	103	Raj	2026-02-01	47000
9	103	Raj	2026-03-01	49000
10	104	Sneha	2026-01-01	70000
11	104	Sneha	2026-02-01	70000
12	104	Sneha	2026-03-01	72000

EmployeeSalaryHistory



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Solution:

employee_id integer	employee_name character varying (50)
103	Raj
101	Alok

(B) A company maintains employee details in an Employee table and department information in a Department table.

Write an SQL query to find the employees whose salaries belong to the **top three distinct salary levels within their respective departments**.

If multiple employees have the same salary, they should receive the same salary rank and all of them should be included in the result.

The output should display:

- Department Name
- Employee Name
- Salary

Department

id	name
1	IT
2	Sales
3	HR

Employee

id	name	salary	departmentId
1	Max	90000	1
2	Jim	85000	1
3	John	85000	1
4	Alex	75000	1
5	Sam	70000	1
6	Henry	80000	2
7	Susan	75000	2
8	David	70000	2
9	Robert	65000	2
10	Tina	60000	3
11	Rita	55000	3
12	Manoj	50000	3



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Solution:

department character varying (50) 🔒	employee character varying (50) 🔒	salary integer 🔒
IT	Max	90000
IT	Jim	85000
IT	John	85000
IT	Alex	75000
Sales	Henry	80000
Sales	Susan	75000
Sales	David	70000
HR	Tina	60000
HR	Rita	55000
HR	Manoj	50000

3. Learning Outcomes:

- Learned to use Window Functions for analyzing employee data.
- Applied functions to compare current and previous salary records.
- Learned to rank employees based on salary within departments.
- Understood how to handle equal salaries using ranking functions.
- Developed practical skills in analyzing salary trends using SQL.